

the complements of their intersection using Demorgans or dualization law is \_\_\_\_\_.

62. In tossing a fair coin turning up of head and tail are \_\_\_\_\_ outcomes.

63. The turning up of spots, 1, 2, 3, 4, 5 and 6 in rolling of a die are \_\_\_\_\_ and \_\_\_\_\_ events.

64. The probability of obtaining a total of 8 in a single throw of two dice is \_\_\_\_\_.

65. If  $A$  is an arbitrary event, then  $P(A/A) =$  \_\_\_\_\_.

66. For any two arbitrary events  $A$  and  $B$ ,  $P(\bar{A}|B) =$  \_\_\_\_\_.

67. For any given event  $C$ ,  $P(\phi|C) =$  \_\_\_\_\_.

68. For any two events  $A$  and  $B$ ,  $P(A \cup \bar{B}) =$  \_\_\_\_\_.

69. For any two events  $A$  and  $B$ ,  $P(\bar{A} \cap \bar{B}) =$  \_\_\_\_\_.

70. If two events  $A$  and  $B$  are mutually exclusive and  $P(A) = \frac{1}{3}$ ,  $P(B) = \frac{1}{2}$ , then

(i) Probability that either  $A$  or  $B$  will occur is \_\_\_\_\_.

(ii) Probability that neither  $A$  nor  $B$  will occur is \_\_\_\_\_.

(iii) Probability that  $A$  and  $B$  both will occur is \_\_\_\_\_.

71. Making use of the multiplication theorem,

the expression for  $P(A \cap B \cap C \cap D)$  is \_\_\_\_\_.

72. If no outcome of an experiment is expected to occur more frequently than others, the outcomes are \_\_\_\_\_.

73. If an event is not simple, it is a \_\_\_\_\_ event.

74. The outcomes of a statistical experiment which result in the happening of an event are called \_\_\_\_\_.

75. If two events are not independent, they are obviously \_\_\_\_\_.

76. Probability interpreted as a measure of degree of belief of an individual is called \_\_\_\_\_.

77. Assigning of equal probabilities to all the elementary events of a statistical experiment is called \_\_\_\_\_ of probabilities.

78. If  $P(A \cap B) = P(A)P(B)$ ,  $P(A \cap C) = P(A)P(C)$  and  $P(B \cap C) = P(B)P(C)$ , the events  $A$ ,  $B$  and  $C$  are \_\_\_\_\_.

79. If a class has 60 per cent boys and 40 per cent girls and the probability of getting first class of a girl is 0.30 and that of a boy getting first class is 0.25, the probability of a randomly chosen student getting first class is \_\_\_\_\_.

80. If  $A$  and  $B$  are two mutually exclusive and exhaustive events and  $P(B) = 2P(A)$ , then  $P(A) =$  \_\_\_\_\_.

## SECTION-C

### Multiple Choice Questions

Select the correct alternative out of given ones:

Q. 1 The outcome of tossing a coin is a:

- (a) simple event
- (b) mutually exclusive event
- (c) complementary event
- (d) compound event

Q. 2 Classical probability is measured in terms of:

- (a) an absolute value
- (b) a ratio
- (c) absolute value and ratio both
- (d) none of the above

Q. 3 Probability can take values

## ELEMENTARY PROBABILITY

(a)  $-\infty$  to  $\infty$

(b)  $-\infty$  to 1

(c)  $-1$  to 1

(d) 0 to 1

Q. 4 Probability is expressed as:

- (a) ratio
- (b) proportion
- (c) percentage
- (d) all the above

Q. 5 Two events are said to be independent if:

- (a) each outcome has equal chance of occurrence
- (b) there is no common point in between them
- (c) one does not affect the occurrence of the other
- (d) both the events have only one point

Q. 6 If  $A$  and  $B$  are two events which have no point in common, the events  $A$  and  $B$  are:

- (a) complementary to each other
- (b) independent
- (c) mutually exclusive
- (d) dependent

Q. 7 Classical probability is also known as:

- (a) Laplace's probability
- (b) mathematical probability
- (c) a priori probability
- (d) all the above

Q. 8 Each outcome of a random experiment is called:

- (a) Primary event
- (b) compound event
- (c) derived event
- (d) all the above

Q. 9 If  $A$  and  $B$  are two events, the probability of occurrence of either  $A$  or  $B$  is given as:

- (a)  $P(A) + P(B)$
- (b)  $P(A \cup B)$
- (c)  $P(A \cap B)$
- (d)  $P(A)P(B)$

Q. 10 If  $A$  and  $B$  are two events, the probability of occurrence of  $A$  and  $B$  simultaneously is given as:

- (a)  $P(A) + P(B)$

(b)  $P(A \cup B)$

(c)  $P(A \cap B)$

(d)  $P(A)P(B)$

Q. 11 The limiting relative frequency approach of probability is known as:

- (a) statistical probability
- (b) classical probability
- (c) mathematical probability
- (d) all the above

Q. 12 The definition of statistical probability was originally given by:

- (a) De Moivre
- (b) Laplace
- (c) Von-Mises
- (d) Pascal

Q. 13 The definition of a priori probability was originally given by:

- (a) De Moivre
- (b) Laplace
- (c) Von-Mises
- (d) Feller

Q. 14 If it is known that an event  $A$  has occurred, the probability of an event  $E$  given  $A$  is called:

- (a) empirical probability
- (b) a priori probability
- (c) posteriori probability
- (d) conditional probability

Q. 15 Probability by classical approach has:

- (a) no lecnuae
- (b) only one lecnua
- (c) only two lecnuae
- (d) many lecnuae

Q. 16 Classical probability is possible in case of:

- (a) unequikely outcomes
- (b) equikely outcomes
- (c) either unequikely or equikely outcomes
- (d) all the above

Q. 17 An event consisting of those elements which are not in  $A$  is called:

- (a) primary event
- (b) derived event
- (c) simple event

- (d) complementary event
- Q. 18 The probability of all possible outcomes of a random experiment is always equal to:
- infinity
  - zero
  - one
  - none of the above
- Q. 19 The probability of the intersection of two mutually exclusive events is always:
- infinity
  - zero
  - one
  - none of the above
- Q. 20 The individual probabilities of occurrence of two events  $A$  and  $B$  are known, the probability of occurrence of both the events together will be:
- increased
  - decreased
  - one
  - zero
- Q. 21 If  $E_1, E_2, \dots, E_n$  is a countable sequence of events such that  $E_i \supset E_{i+1}$  for  $i = 1, 2, \dots$  then:
- $\lim_{n \rightarrow \infty} P(E_n) = 0$
  - $\lim_{n \rightarrow \infty} P(E_n) = \infty$
  - $\lim_{n \rightarrow \infty} P(E_n) = 1$
  - $\lim_{n \rightarrow \infty} P(E_n) = \text{impossible value}$
- Q. 22 If  $A_1, A_2$  and  $A_3$  are three mutually exclusive events, the probability of their union is equal to:
- $P(A_1)P(A_2)P(A_3)$
  - $P(A_1) + P(A_2) + P(A_3) - P(A_1 A_2 A_3)$
  - $P(A_1) + P(A_2) + P(A_3)$
  - $P(A_1)P(A_2) + P(A_1) \times P(A_3) + P(A_2)P(A_3)$
- Q. 23 If  $A_1, A_2$  and  $A_3$  are three independent events, the probability of their joint occurrence is equal to:
- $P(A_1)P(A_2)P(A_3)$
  - $1/P(A_1)P(A_2)P(A_3)$
  - $P(A_1) + P(A_2) + P(A_3)$
  - $P(A_1 \cap A_2) + P(A_1 \cap A_3) + P(A_2 \cap A_3)$
- Q. 24 If two events  $A$  and  $B$  are such that  $A \subset B$  and  $B \subset A$ , the relation between  $P(A)$  and  $P(B)$  is:
- $P(A) \leq P(B)$
  - $P(A) \geq P(B)$
  - $P(A) = P(B)$
  - none of the above
- Q. 25 If  $A$  is an event, the conditional probability of  $A$  given  $A$  is equal to:
- zero
  - one
  - infinity
  - indeterminate quantity
- Q. 26 If  $A \subset B$ , the probability,  $P(A/B)$  is equal to:
- zero
  - one
  - $P(A)/P(B)$
  - $P(B)/P(A)$
- Q. 27 If  $B \subset A$ , the probability  $P(A/B)$  is equal to:
- zero
  - one
  - $P(A)/P(B)$
  - $P(B)/P(A)$
- Q. 28 If two events  $A$  and  $B$  are such that  $A \subset B$ , the relation between the conditional probabilities  $P(A/C)$  and  $P(B/C)$  is:
- $P(A/C) = P(B/C)$
  - $P(A/C) > P(B/C)$
  - $P(A/C) < P(B/C)$
  - all the above

## ELEMENTARY PROBABILITY

- Q. 29 For any two events  $A$  and  $B$ ,  $P(A - B)$  is equal to:
- $P(A) - P(B)$
  - $P(B) - P(A)$
  - $P(B) - P(AB)$
  - $P(A) - P(AB)$
- Q. 30 If an event  $B$  has occurred and it is known that  $P(B) = 1$ , the conditional probability  $P(A/B)$  is equal to:
- $P(A)$
  - $P(B)$
  - one
  - zero
- Q. 31 If  $A$  and  $B$  are any two mutually exclusive events, the  $P(A/A \cup B)$  is equal to:
- $P(A)/[P(A) + P(B)]$
  - $P(A \cup B)/[P(A) + P(B)]$
  - $P(B)/P(A \cup B)$
  - none of the three
- Q. 32 If  $A$  and  $B$  are two independent events, then  $P(\bar{A} \cap \bar{B})$  is equal to:
- $P(\bar{A})P(\bar{B})$
  - $1 - P(A \cup B)$
  - $[1 - P(A)][1 - P(B)]$
  - all the above
- Q. 33 If  $E_1, E_2, \dots, E_n$  are  $n$  mutually exclusive events such that  $P(E_j) \neq 0$  for  $j = 1, 2, \dots, n$  and  $A$  is an arbitrary event contained in  $\cup E_j$  with  $P(A) > 0$  and has been observed, the probability of a particular event  $E_j$  given  $A$  is given by the formula
- $P(E_j|A) = \frac{P(E_j)P(A|E_j)}{\sum_j P(E_j)}$
  - $P(E_j|A) = \frac{P(E_j)P(A|E_j)}{\sum_j P(E_j)P(A|E_j)}$
  - $P(E_j|A) = \frac{P(A|E_j)}{\sum_j P(A|E_j)}$
  - none of the above
- Q. 34 If  $A$  and  $B$  are two events such that  $AB$  and  $\bar{A}\bar{B}$  are two mutually exclusive and exhaustive events in which the event  $A$  can occur, then
- $P(A) = 1$
  - $P(A) = P(AB) + P(\bar{A}\bar{B})$
  - $P(A) = P(\bar{A}) + P(\bar{A}\bar{B})$
  - $P(A) = P(\bar{A}B) + P(\bar{A}\bar{B})$
- Q. 35 If  $k$  toffees are distributed at random among  $n$  children, the probability that a child will receive exactly  $r$  toffees is:
- $\frac{kc_r(n-1)^{n-r}}{n^k}$
  - $\frac{kc_r(k-1)^{k-r}}{k^n}$
  - $\frac{kc_r(n-1)^r}{n^k}$
  - $\frac{kc_r(n-1)^{n-r}}{k^n}$
- Q. 36 The idea of posteriori probabilities was introduced by
- Pascal
  - Peter and Paul
  - Thomas Bayes
  - M. Loe've
- Q. 37 In a city 60 per cent read newspaper  $A$ , 40 per cent read newspaper  $B$  and 30 per cent read newspaper  $C$ , 20 per cent read  $A$  and  $B$ , 30 per cent read  $A$  and  $C$ , 10 per cent read  $B$  and  $C$ . Also 15 per cent read papers  $A, B$  and  $C$ . The percentage of people who do not read any of these newspapers is:
- 65 per cent
  - 15 per cent
  - 45 per cent
  - none of the above
- Q. 38 If a bag contains 4 white and 3 black balls. Two draws of 2 balls are successively made, the probability of getting 2 white balls at

first draw and 2 black balls at second draw when the balls drawn at first draw were replaced is:

- (a)  $3/7$
- (b)  $1/7$
- (c)  $19/49$
- (d)  $2/49$

Q. 39 In question 38, if the balls are not replaced after the first draw, the probability of 2 white balls at first draw and 2 black balls at second draw is:

- (a)  $\frac{3}{35}$
- (b)  $\frac{13}{35}$
- (c)  $\frac{1}{2}$
- (d) none of the above

Q. 40 In tossing three coins at a time, the probability of getting at most one head is:

- (a)  $\frac{3}{8}$
- (b)  $7/8$
- (c)  $1/2$
- (d)  $1/8$

Q. 41 There is 80 per cent chance that a problem will be solved by a statistics student and 60 per cent chance is there that the same problem will be solved by the mathematics student. The probability that at least the problem will be solved is:

- (a) 0.48
- (b) 0.92
- (c) 0.10
- (d) 0.75

Q. 42 The probability of two persons being born on the same day (ignoring date) is:

- (a)  $1/49$
- (b)  $1/365$
- (c)  $1/7$
- (d) none of the above

Q. 43 An urn contains 5 red, 4 white and 3 black

balls. The probability of three balls being of different colours when the ball is replaced after each draw is equal to:

- (a)  $3/144$
- (b)  $4/144$
- (c)  $5/144$
- (d) 1

Q. 44 In question 43, the probability of three balls being drawn in the order red, white and black when the balls are not replaced after each draw, is equal to:

- (a)  $1/22$
- (b)  $5/144$
- (c)  $60/144$
- (d) none of the above

Q. 45 An urn A contains 5 white and 3 black balls and B contains 4 white and 4 black balls. An urn is selected and a ball is drawn from it, the probability, that the ball is white, is:

- (a)  $9/8$
- (b)  $9/16$
- (c)  $5/32$
- (d)  $5/16$

Q. 46 From a pack of 52 cards, two cards are drawn at random. The probability that one is an ace and the other is a king is:

- (a)  $2/13$
- (b)  $1/169$
- (c)  $16/169$
- (d)  $8/663$

Q. 47 Two dice are rolled by two players A and B. A throws 10, the probability that B throws more than A is:

- (a)  $1/12$
- (b)  $1/6$
- (c)  $1/18$
- (d) none of the above

Q. 48 The data reveals that 10 per cent patients die in a particular type of operation. A doctor performed 9 operations and all of them survived. Whether the 10th patient on being operated:

- (a) will survive
- (b) will die

- (c) may survive or may die
- (d) none of the above

Q. 49 There are two groups of students consisting of 4 boys and 2 girls; 3 boys and 1 girl. One student is selected from both the groups. The probability of one boy and one girl being selected is:

- (a)  $1/9$
- (b)  $\frac{5}{12}$
- (c) 1
- (d) none of the above

Q. 50 In a shooting competition, Mr X can shoot at the bulls eye 4 times out of 5 shots and Mr Y, 5 times out of six and Mr Z, 3 times out of 4 shots. The probability that the target will be hit at least twice is:

- (a)  $107/120$
- (b)  $47/120$
- (c)  $1/2$
- (d) none of the above

Q. 51 There are two bags. One bag contains 4 red and 5 black balls and the other 5 red and 4 black balls. One ball is to be drawn from either of the two bags. the probability of drawing a black ball is:

- (a) 1
- (b)  $16/81$
- (c)  $1/2$
- (d)  $10/81$

Q. 52 Three dice are rolled simultaneously. The probability of getting 12 spots is:

- (a)  $1/8$
- (b)  $25/216$
- (c)  $1/12$
- (d) none of the above

Q. 53 Given that  $P(A) = \frac{1}{3}$ ,  $P(B) = \frac{1}{4}$ ,  $P(A|B)$

$= \frac{1}{6}$ , the probability  $P(B|A)$  is equal to:

- (a)  $1/4$
- (b)  $3/4$
- (c)  $1/8$
- (d) none of the above

Q. 54 From the probabilities given in question 53, the probability,  $P(B|\bar{A})$  is equal to:

- (a)  $1/16$
- (b)  $15/24$
- (c)  $15/16$
- (d)  $5/16$

Q. 55 A number is selected randomly from each of the two sets

- 1, 2, 3, 4, 5, 6, 7, 8
- 2, 3, 4, 5, 6, 7, 8, 9

The probability that the sum of the numbers is equal to 9 is:

- (a)  $8/91$
- (b)  $7/72$
- (c)  $14/81$
- (d)  $7/64$

Q. 56 A bag contains 3 white and 5 red balls. Three balls are drawn after shaking the bag. The odds against these balls being red is:

- (a)  $5/28$
- (b)  $5/8$
- (c)  $15/64$
- (d)  $3/5$

Q. 57 A bag contains 3 white, 1 black and 3 red balls. Two balls are drawn from the well shaken bag. The probability of both the balls being black is:

- (a) 1
- (b) zero
- (c)  $1/7$
- (d) none of the above

Q. 58 The chance of winning the race of the horse

A in Durby is  $\frac{1}{5}$  and that of horse B is  $\frac{1}{6}$ .

The probability that the race will be won by A or B is:

- (a)  $1/30$
- (b)  $1/3$
- (c)  $11/30$
- (d) none of the above

Q. 59 Four cards are drawn from a pack of 52 cards. The probability that out of 4 cards being 2 red and 2 black is:

- Q. 81 *A* can hit a target 2 times in 5 shots, *B* 2 times in 5 shots and *C* 4 times in 5 shots. They fire a volley (each try once to hit the target). The probability that two shots hit is:
- 24/125
  - 67/125
  - 121/125
  - 58/125
- Q. 82 There are four coins in a bag. One of the coins has head on both sides. A coin is drawn at random and tossed five times and fell always with head upward. The probability that it is the coin with two head is:
- 3/128
  - 1/4
  - 32/35
  - none of the above
- Q. 83 One of the two events is certain to happen. The chance of one event is one-fifth of the other. The odds in favour of the other is:
- 1 : 6
  - 6 : 1
  - 5 : 1
  - 1 : 5
- Q. 84 One of the two events must happen; given that the chance of one is one-fourth of the other. The odd in favour of the other is:
- 1 : 3
  - 1 : 4
  - 1 : 5
  - none of the above
- Q. 85 A coin is tossed six times. The probability of obtaining heads and tails alternately is:
- 1/64
  - 1/2
  - 1/32
  - none of the above
- Q. 86 The odds in favour of certain event are 5 : 4, and odds against another event are 4 : 3. The chance that at least one of them will happen is:
- 15/63
  - 51/63
  - 47/63
  - none of the above
- Q. 87 *A* and *B* start in a ring with ten other persons. If the arrangement of 12 persons is at random, the chance that there are exactly three persons between *A* and *B* is:
- 1/66
  - 2/11
  - 4/11
  - none of the above
- Q. 88 Three houses were available in a locality for allotment. Three persons applied for a house. The probability that all the three persons applied for the same house is:
- 1/3
  - 1/9
  - 1/27
  - 1
- Q. 89 In the problem of question 88, the probability that each of the three applied for a different house is:
- 1/9
  - 1/27
  - 1
  - 2/9
- Q. 90 *A* speaks truth 4 times out of five and *B* speaks truth 3 times out of four. They agree in the assertion that a white ball has been drawn from a bag containing 10 balls of different colours. The probability that a white ball was really drawn is:
- 3/50
  - 1/27
  - 1/1350
  - 81/82
- Q. 91 In the problem of question 90 if the bag contains 1 white and 9 red balls, the probability of one white ball being drawn is:
- 4/7
  - 3/50
  - 9/200
  - none of the above
- Q. 92 If *A* tells truth 4 times out of 5 and *B* tells truth 3 times out of 4. The probability that both expressing the same fact contradict each other is:
- 1/20

## ELEMENTARY PROBABILITY

- 3/20
  - 1/5
  - none of the above
- Q. 93 The probability of drawing a white ball in the first draw and again a white ball in the second draw with replacement from a bag containing 6 white and 4 blue balls is:
- 2/10
  - 6/10
  - 36/100
  - 1/3
- Q. 94 A fair coin is tossed repeatedly unless a head is obtained. The probability that the coin has to be tossed at least four times is:
- 1/2
  - 1/4
  - 1/6
  - 1/8
- Q. 95 Out of 20 employees in a company, five are graduates. Three employees are selected at random. The probability of all the three being graduates is:
- 1/64
  - 1/125
  - 1/114
  - none of the above
- Q. 96 In the problem of question 95, the probability that at least one of them is a graduate is:
- 11/114
  - 137/1368
  - 137/228
  - none of the above
- Q. 97 (Parzen) In answering a question on a multiple choice test, an examinee either knows the answer with probability  $p$  or he guesses with probability  $(1 - p)$ . Let the probability of answering the question correctly be 1 for an examinee who knows the answer and  $\frac{1}{m}$  who guesses ( $m$  being the number of multiple choice alternatives). Suppose an examinee answer a question correctly. The probability that he really knows the answer is:
- $\frac{mp}{1 + mp}$
  - $\frac{mp}{1 + (m - 1)p}$
  - $\frac{(m - 1)p}{1 + (m - 1)p}$
  - none of the above
- Q. 98 A machine part is produced by three factories *A*, *B* and *C*. Their proportional production is 25, 35 and 40 per cent, respectively. Also, the percentage defectives manufactured by three factories are 5, 4 and 3, respectively. A part is taken at random and is found to be defective. The probability that the selected part belongs to factor *B* is:
- 28/503
  - 4/11
  - 14/276
  - none of the above
- Q. 99 A card is drawn from a well shuffled pack of 52 cards. A gambler bets that it is either a heart or an ace. What are odds against his winning this bet?
- 9 : 4
  - 4 : 9
  - 35 : 52
  - 1 : 3
- Q. 100 If one card is selected at random from 100 cards numbered as 00, 01, ..., 99. Suppose  $x$  and  $y$  are the sum and product of the digits on the selected card. If  $i$  is a whole number, the probability  $P(x = iy, i \neq 0)$  is equal to:
- $\frac{1}{19}$
  - 1/50
  - 19/100
  - none of the above
- Q. 101 Two dice, each numbered 1 to 6, are thrown together. Consider two events *A* and *B* given by
- A* - even number of the first die

$B$  - number on the second die is greater than 4.

Then  $P(A \cup B)$  is:

- (a)  $2/3$
- (b)  $1/6$
- (c)  $1/2$
- (d)  $3/4$

**Q. 102** An unbiased coin is tossed four times. The probability that the number of heads exceeds the number of tails is:

- (a)  $1/12$
- (b)  $3/4$
- (c)  $3/8$
- (d)  $5/16$

**Q. 103** If  $P(A|B) = \frac{1}{4}$  and  $P(B|A) = \frac{1}{3}$ , then  $P(A)/P(B)$  is equal to:

- (a)  $3/4$
- (b)  $7/12$
- (c)  $4/3$
- (d)  $1/12$

**Q. 104** For two events  $E_1, E_2$ , if  $P(E_1) = 1/2$ ,  $P(E_2) = \frac{1}{3}$ ,  $P(E_1 \cup E_2) = \frac{2}{3}$ , then  $P(E_1 \cap E_2)$  is equal to:

- (a)  $1/4$
- (b)  $1/6$
- (c)  $2/5$
- (d)  $1/3$

**Q. 105** For two events  $A_1$  and  $A_2$ , if  $P(A_1) = \frac{2}{3}$ ,

$P(A_2) = \frac{3}{8}$  and  $P(A_1 \cap A_2) = \frac{1}{4}$ , then  $A_1$  and  $A_2$  are:

- (a) mutually exclusive but not independent
- (b) mutually exclusive and independent
- (c) independent but not mutually exclusive
- (d) not mutually exclusive and not independent

**Q. 106.** In a library there are 40 per cent mathematics books and remaining 60 per cent science books. It is known that 2 per cent of the mathematics books are in Hindi and 1 per

cent of science books are in Hindi. If one book is taken out at random and is found to be in Hindi, the probability that it is a science book is:

- (a)  $2/9$
- (b)  $3/7$
- (c)  $6/13$
- (d)  $1/4$

**Q. 107** Take four identical marbles. On the first write symbols  $A_1, A_2, A_3$ . On each of the other three write  $A_1, A_2, A_3$  respectively. Put the four marbles in an urn and draw one at random. Let  $E_i$  denotes the event that the symbol  $A_i$  appears on the drawn marble. Then which of the four statements is true for this problem?

- (a)  $P(E_1) = P(E_2) = P(E_3) = \frac{1}{4}$
- (b)  $P(E_1 E_2) = P(E_2 E_3) = P(E_1 E_3) = \frac{1}{4}$
- (c)  $P(E_1 E_2 E_3) = \frac{1}{8}$
- (d) all the above.

**Q. 108** For a post in a factory, husband and wife both applied. The probability of selection of a male is  $\frac{1}{5}$  and that of a female is  $\frac{1}{3}$ . The probability of selection of only one of them is:

- (a)  $2/15$
- (b)  $4/15$
- (c)  $8/15$
- (d)  $2/3$

**Q. 109** Consider a family of three children. Also assume that all possible distributions of children have same probabilities. Let  $H$  be the event, 'the family has children of both the sexes' and  $A$  be the event, 'there is at most one girl'. Then the events  $H$  and  $A$  are:

- (a) mutually exclusive
- (b) complementary events
- (c) independent events
- (d) exhaustive events.

**Q. 110** In a certain residential suburb, 70 per cent of all households subscribe to the metropolitan newspaper, 60 per cent subscribe to the local evening paper and 40 per cent households subscribe to both the papers. The probability that a household selected at random subscribes to one of the two newspapers is:

- (a) 0.3
- (b) 0.5
- (c) 0.9
- (d) 0.2

## ANSWERS

### SECTION-B

(1) Laplace (2) event (3) sets (4) compound (5) no common point (6) null (7) elementary (8) infinite (9) not equally likely (10) a priori (11) total (12) null or impossible (13)  $A \subset B$  and  $B \subset A$  (14) exhaustive (15) union (16) intersection (17) Von Mises (18) statistical or empirical (19) infinite (20) derived (21) sample space (22) zero (23) one (24) one (25) negative (26) 0 to 1 (27) gambling (28) product (29) same sample space (30) ratio or fraction or percentage (31) independent (32) exhaustive (33)  $P(A) \geq P(B)$  (34)  $P(A) - P(B)$  or  $P(A - B)$  (35)  $P(A) - P(A \cap B)$  (36)  $P(B) - P(A \cap B)$  (37) additive (38)  $P(A) + P(B)$  (39)  $P(A) + P(B) + P(C) - P(A \cap C) - P(B \cap C)$  (40)  $P(A) \times P(B)$  (41)  $P(B)$  (42)  $P(A|C) + P(B|C) - P(AB|C)$  (43)  $P(A|C)$  (44)  $P(A)P(B|A)$  or  $P(B)P(A|B)$  (45)  $P(A)$  (46) independent (47)  $P(A)P(B)P(C|A \cap B)$  or  $P(A)P(C)P(B|A \cap C)$  or  $P(B)P(C)P(A|B \cap C)$  (48)  $P(A)P(C)$  (49) independent (50)  $\leq$  (51)  $P(A) + P(B) - 2P(A \cap B)$  (52) (a)  $1 - p_3$  (b)  $p_2 - p_3$  (c)

$1 - p_1 - p_2 + p_3$  (d)  $1 - p_1 + p_2$  (e)  $\frac{p_3}{p_2}$  (53) 1

(54) inverse (55) posteriori (56)  $P(A_1|B) + P(A_2|B) - P(A_1 A_2|B)$  (57)  $[P(A) - P(AB)]/[1 - P(B)]$  (58)  $P(A_1)/[P(A_1) + P(A_2)]$  (59)  $P(A)$

(60)  $P(\bar{A} \cap \bar{B})$  (61)  $P(\bar{A} \cup \bar{B})$  (62) exhaustive

(63) mutually exclusive; exhaustive (64)  $\frac{5}{36}$  (65) (66)  $1 - P(A/B)$  (67) zero (68)  $1 - P(A \cap B)$  (69)  $1 - P(A \cup B)$  (70) (i)  $5/6$  (ii)  $1/6$  (iii) 0 (71)  $P(A)P(B|A)P(C|A \cap B)P(D|A \cap B \cap C)$  (72) equally likely. (73) Compound (74) favourable cases (75) dependent (76) subjective probability (77) natural assignment (78) pairwise independent (79) 0.27 (80)  $1/3$

## SECTION-C

### Answers

(1) a (2) b (3) d (4) d (5) c (6) c (7) d (8) a (9) b (10) c (11) a (12) c (13) b (14) d (15) d (16) b (17) d (18) c (19) b (20) b (21) a (22) c (23) a (24) c (25) b (26) c (27) b (28) c (29) d (30) a (31) a (32) d (33) b (34) b (35) a (36) c (37) b [Hint:  $P(A \cup B \cup C) = 0.85$ , and required percentage =  $100 - 85 = 15$  per cent]

(38) d [Hint:  $P(2W) = \frac{4C_2}{7C_2} = \frac{2}{7}$ ,  $P(2B) = \frac{3C_2}{7C_2} = \frac{3}{7}$

$$\frac{1}{7}, P(2W \text{ and } 2B) = \frac{2}{7} \times \frac{1}{7} = \frac{2}{49}]$$

(39) a [Hint:  $P(2W) = \frac{2}{7}$ ,  $P(2B) = \frac{3C_2}{5C_2} = \frac{3}{10}$ ,

$$P(2W \text{ and } 2B) = \frac{2}{7} \times \frac{3}{10} = \frac{3}{35}]$$

(40) c [Hint:  $\Omega = 8$  points,  $E = \text{HTT, THT, TTH, TTT}$ ,  $P(E) = \frac{4}{8} = \frac{1}{2}$ ]

(41) b [Hint  $P(E) = 0.8 + 0.6 - 0.8 \times 0.6 = 0.92$ ]

(42) c [Hint:  $P(E) = 7\left(\frac{1}{7} \times \frac{1}{7}\right) = \frac{1}{7}$ ]

(43) c [Hint:  $P(E) = \frac{5}{12} \times \frac{4}{12} \times \frac{3}{12} = \frac{5}{144}$ ]

$$(44) a \left[ \text{Hint: } P(E) = \frac{5}{12} \times \frac{4}{11} \times \frac{3}{10} = \frac{1}{22} \right]$$

$$(45) d \left[ \text{Hint: } P(E) = \frac{1}{2} \left( \frac{5}{8} + \frac{4}{8} \right) = \frac{9}{16} \right]$$

$$(46) d \left[ \text{Hint: } P(E) = \frac{4C_1 \times 4C_1}{52C_2} = \frac{8}{663} \right]$$

$$(47) a \left[ \text{Hint: } \Omega = 36 \text{ points, } E: (6, 6), (6, 5), (5, 6), \right. \\ \left. P(E) = \frac{3}{36} = \frac{1}{12} \right]$$

(48) c

$$(49) b \left[ \text{Hint: } P(E) = \left( \frac{4c_1 \times 1c_1}{6c_1 \times 4c_1} \right) + \left( \frac{2c_1}{6c_1} \times \frac{3c_1}{4c_1} \right) \right. \\ \left. = \frac{5}{12} \right]$$

$$(50) a \left[ \text{Hint: } P(E) = \frac{4}{5} \times \frac{5}{6} \times \frac{1}{4} + \frac{4}{5} \times \frac{1}{6} \times \frac{3}{4} + \frac{1}{5} \times \frac{5}{6} \right. \\ \left. \times \frac{3}{4} + \frac{4}{5} \times \frac{5}{6} + \frac{3}{4} = \frac{107}{120} \right]$$

$$(51) c \left[ \text{Hint: } P(E) = \frac{1}{2} \left( \frac{5}{9} + \frac{4}{9} \right) = \frac{1}{2} \right]$$

$$(52) b \left[ \text{Hint: } \Omega = 6 \times 6 \times 6 \text{ points, } E = (1, 5, 6), \right. \\ (1, 6, 5) \dots, (6, 4, 2), (6, 5, 1) = 25 \text{ points, } P(E) \\ \left. = \frac{25}{216} \right]$$

$$(53) c \left[ \text{Hint: } P(B/A) = \frac{P(A/B)P(B)}{P(A)} = \frac{1}{8} \right]$$

$$(54) d \left[ \text{Hint: } P(B/\bar{A}) = \frac{[1 - P(A/B)]P(B)}{1 - P(A)} = \frac{5}{16} \right]$$

$$(55) d \left[ \text{Hint: } \Omega = 54 \text{ points, } E = (1, 8), (2, 7), \right. \\ (3, 6), (4, 5), (5, 4), (6, 3), (7, 2), P(E) = \frac{7}{64} \left. \right]$$

$$(56) a \left[ \text{Hint: } P(E) = \frac{5C_3}{8C_3} = \frac{5}{28} \right]$$

(57) b [Hint: Impossible event]

$$(58) c \left[ \text{Hint: } P(E) = \left( \frac{1}{5} + \frac{1}{6} \right) = \frac{11}{30} \right]$$

$$(59) a \left[ \text{Hint: } P(E) = \frac{26c_2 \times 26c_2}{52c_4} = \frac{325}{833} \right]$$

$$(60) d \left[ \text{Hint: } P(E) = \frac{1}{5} + \frac{1}{7} - \frac{1}{5} \times \frac{1}{7} = \frac{11}{35} \right]$$

$$(61) b \left[ \text{Hint: } P(E) = \frac{5}{12} + \frac{2}{7} - \frac{5}{12} \times \frac{2}{7} = \frac{49}{84} \right]$$

$$(62) d \left[ \text{Hint: } P(E) = \frac{7}{13} + \frac{5}{8} - \frac{7}{13} \times \frac{5}{8} = \frac{43}{52} \right]$$

$$(63) b \left[ \text{Hint: } P(E) = \frac{1}{3} + \frac{1}{5} = \frac{8}{15} \right]$$

$$(64) d \left[ \text{Hint: } \Omega = 36 \text{ points, } E = 18 \text{ pairs of odd} \right. \\ \left. \text{sum, } P(E) = \frac{18}{36} = \frac{1}{2} \right]$$

$$(65) c \left[ \text{Hint: } \Omega 36, \text{ pairs } E = \text{first throw 1 and second} \right. \\ \text{throw 2, 3, 4, 5, 6 or 1st throw 2, 3, 4, 5, 6 and 2nd} \\ \text{throw 1 or both the throws 1, 1. } E = 11 \text{ points } P(E) \\ \left. = 11/36 \right]$$

$$(66) b \left[ \text{Hint: } P(E) = \frac{8}{9} + \frac{6}{7} - \frac{8}{9} \times \frac{6}{7} = \frac{62}{63} \right]$$

$$(67) d \left[ \text{Hint: } P(E) = \frac{4c_2 \times 3c_1}{9c_3} + \frac{4c_2 \times 2c_1}{9c_3} = \frac{5}{14} \right]$$

## ELEMENTARY PROBABILITY

(68) b [Hint: A leap year has 366 days, i.e., 52 week and 2 days. 52 sundays are sure. 53rd sunday may be on the remaining two day. The possible pairs of days are (S, M), (M, T), (T, W), (W, Th), (Th, F) (F, Sat), (Sat, S).  $E = (S, M)$  and  $(Sat, S)$ .  $P(E) = 2/7$ ]

(69) d [Hint:  $\Omega = 36$  pairs,  $E = (4, 6)$  (6, 4), (5, 5), (5, 6), (6, 5) and (6, 6).  $P(E) = 6/36 = 1/6$ ]

$$(70) c \left[ \text{Hint: } P(E) = \frac{3}{4} \times \frac{4}{5} \times \frac{1}{6} + \frac{3}{4} \times \frac{1}{5} \times \frac{5}{6} + \frac{1}{4} \right. \\ \left. \times \frac{4}{5} \times \frac{5}{6} = \frac{47}{120} \right]$$

(71) b [Hint: A - 3 defectives out of 1st eight examined record players B - 9th piece is defective.

$$P(A) = \frac{\binom{4}{3} \binom{11}{5}}{\binom{15}{8}}, P(B/A) = \frac{1}{7}, P(A \cap B) =$$

$$P(B/A)P(A) = 8/195$$

(72) d [Hint:  $E_1$ : disease X is correctly diagnosed by doctor A,  $E_2$ : Patient having disease X dies.

$$P(E_1) = 0.6, P(\bar{E}_1) = 0.4, P(E_2/E_1) = 0.4, \\ P(E_2/\bar{E}_1) = 0.7$$

$$P(E_1/E_2) = \frac{P(E_2/E_1)P(E_1)}{P(E_2/E_1)P(E_1) + P(E_2/\bar{E}_1)P(\bar{E}_1)} = \frac{6}{13}$$

$$(73) d \left[ \text{Hint: } P(A_1) = P(A_2) = P(A_3) = \frac{1}{2}, P \right.$$

$$(A_1 \cap A_2) = \frac{1}{4} = P(A_1)P(A_2). \text{ Similarly, } P(A_2$$

$$\cap A_3) = P(A_2)P(A_3) \text{ and } P(A_1 \cap A_3) = P(A_1)P(A_3)$$

$$\text{Also } P(A_1 \cap A_2 \cap A_3) = 0]$$

(74) d

$$(75) b \left[ \text{Hint: } P(E) = \frac{\binom{8}{3}}{\binom{12}{3}} = \frac{14}{55} \right]$$

$$(76) a \left[ \text{Hint: } P(E_1 E_2 E_3) = P(E_1)P(E_2/E_1) \right.$$

$$\left. P(E_3/E_1 E_2) = \frac{3}{8} \times \frac{5}{10} \times \frac{7}{12} = \frac{7}{64} \right]$$

$$(77) c \left[ \text{Hint: } P(A \cap B) = \frac{1}{3} + \frac{3}{4} - \frac{11}{12} = \frac{1}{6}, P(B/A) \right. \\ \left. = \frac{1/6}{1/3} = \frac{1}{2} \right]$$

$$(78) b \left[ \text{Hint: } P(B'/C') = \frac{P(B' \cap C')}{P(C')} = \frac{P(B' \cup C')}{P(C')} \right.$$

$$\left. = \frac{0.6 + 0.5 - 0.7}{1 - 0.5} = \frac{4}{5} \right]$$

(79) a [Hint:  $P(AB') = P(A) - P(AB) = 0.1$ ,  $P(A/B) = 0.1/0.4 = 1/4$ ]

(80) c [Hint: There are ten digits from 0 to 9 in which a product can end. Prob. of any given digit to

be the first digits is  $\frac{4}{10} = \frac{2}{5}$  for the given event,

$$P(E) = \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} = \left( \frac{2}{5} \right)^4$$

$$(81) d \left[ \text{Hint: } P(E) = \frac{2}{5} \times \frac{3}{5} \times \frac{1}{5} + \frac{2}{5} \times \frac{4}{5} \times \frac{2}{5} + \frac{3}{5} \right.$$

$$\left. \times \frac{4}{5} = \frac{58}{125} \right]$$

(82) c [Hint: Prob. of selecting a coin having both head = 1/4 and it has all five heads =  $\frac{1}{4} \times 1 \times 1 \times 1 \times$

$1 \times 1 = \frac{1}{4}$ . Prob of selection of a coin having head

and tail = 3/4 and it fall with five heads =  $\frac{3}{4} \times \left( \frac{1}{2} \right)^5 =$

$$\frac{3}{128}. P(E) = \frac{1/4}{1/4 + \frac{3}{128}} = \frac{32}{35}$$

$$(83) d \left[ \text{Hint: } p_1 = x, p_2 = \frac{x}{5}, x + \frac{x}{5} = 1, x = \frac{5}{6}, p_2/p_1 = \frac{1/6}{5/6} \text{ or } p_2:p_1 = 1:5 \right]$$

$$P(\text{HTHTHT}) + P(\text{THTHTH}) = \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 = \frac{1}{32}$$

$$(86) c \left[ \text{Hint: } P(E_1) = \frac{5}{9}, P(\bar{E}_2) = \frac{3}{7}, P(E_1 \cup \bar{E}_2) = \frac{5}{9} + \frac{3}{7} - \frac{5}{9} \times \frac{3}{7} = \frac{47}{63} \right]$$

(87) b [Hint: Let A be at 12<sup>th</sup> place, P(A) = 1/12. B can be at 4<sup>th</sup> or 8<sup>th</sup> places out of remaining 11 places.

$$P(B) = \frac{2}{11}. \text{ Also A can be at any of the 12 places.}$$

$$\text{Therefore, } P(E) = \frac{1}{12} \times \frac{2}{11} \times 12 = \frac{2}{11}$$

$$(88) d \left[ \text{Hint: } P(E) = 3 \times \left(\frac{1}{3}\right)^3 = \frac{1}{9} \right]$$

$$(89) d \left[ \text{Hint: } P(E) = \frac{3P_3}{3^3} = \frac{2}{9} \right]$$

(90) d [Hint: Prob. of their true assertion that a white ball is drawn = 3/50. Chance of A statement

being wrong =  $\frac{1}{9} \times \frac{1}{5} = \frac{1}{45}$  and that of B being

wrong =  $\frac{1}{9} \times \frac{1}{3} = \frac{1}{27}$ . Prob. of A and B to agree in

false statement that a white ball is drawn =  $\frac{1}{45} \times \frac{1}{27}$

$$\times \frac{9}{10} = \frac{1}{27 \times 50}. P(E) = \frac{3/50}{3/50 + \frac{1}{27 \times 50}} = \frac{81}{82}$$

$$(91) a \left[ \text{Hint: } P(E) = \frac{3/50}{\frac{3}{5} + \frac{9}{10} \times \frac{1}{5} \times \frac{1}{9}} = \frac{4}{7} \right]$$

$$(92) d \left[ \text{Hint: } P(E) = \frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4} = \frac{7}{20} \right]$$

$$(93) c \left[ \text{Hint: } P(E) = \frac{6c_1 \times 6c_1}{10c_1 \times 10c_1} = \frac{36}{100} \right]$$

$$(94) b \left[ \text{Hint: } P(E) = P(\text{TTHH}) + P(\text{TTTH}) + P(\text{TTTTTH}) + \dots = \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots = \frac{1/8}{1-1/2} = \frac{1}{4} \right]$$

$$(95) c \left[ \text{Hint: } P(E) = 5c_3/20c_3 = 1/114 \right]$$

$$(96) c \left[ \text{Hint: } P(E) = \frac{1}{20c_3} [5c_1 \times 15c_2 + 5c_2 \times 15c_1 + 5c_3] = \frac{137}{228} \right]$$

(97) b [Hint: A<sub>1</sub> - Examinee knows the answer, A<sub>2</sub> - Examinee guesses the answer, B - Answers correctly.

$$P(A_1/B) = \frac{P(A_1) P(B/A_1)}{P(A_1) P(B/A_1) + P(A_2) P(B/A_2)} = \frac{p \times 1}{p \times 1 + \frac{1-p}{m}}$$

$$(98) b \left[ \text{Hint: } P(E) = \right]$$

$$= \frac{35/100 \times 4/100}{\frac{35}{100} \times \frac{4}{100} + \frac{25}{100} \times \frac{5}{100} + \frac{40}{100} \times \frac{3}{100}} = \frac{4}{11}$$

(99) a [Hint: A - card is heart, B - card is ace

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{13}, P(A \cap B) = \frac{1}{52}.$$

$$P(A \cup B) = \frac{4}{13} = p, q = \frac{9}{13}, q:p = 9:4]$$

(100) c [Hint: E = 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 20, 30, 40, 50, 60, 70, 80, 90 = 19 points Ω = 00, 01, ..., 99 = 100 points, P(E) = 19/100]

(101) b (102) d (103) a (104) b (105) c (106) b

(107) b (108) d (109) c (110) c

## Suggested Reading

1. Agarwal, B.L., *Basic Statistics*, New Age International (P) Ltd. Publishers, New Delhi, 3rd edn., 1996.
2. Bhat, B.R., *Modern Probability Theory*, Wiley Eastern Ltd., Publishers, New Delhi, 2nd edn., 1995.
3. Chung, K.L., *Elementary Probability Theory with Stochastic Processes*, Springer Verlag, Berlin, 1975.
4. Feller, W., *An Introduction to Probability Theory and its Applications*, Vol. I & II, John Wiley & Sons, New York, 1966.
5. Finetti, B. D., *Theory of Probability*, Translated by Antonio Machi and Adrian Smith, John Wiley & Sons, New York, 1974.
6. Fish, M., *Probability Theory and Mathematical Statistics*, John Wiley & Son, New York, 1963.
7. Gottinger, H.W., *Elements of Statistical Analysis*, Walter de Gruyter, Berlin, 1980.

8. Haight, F.A., *Applied Probability*, Plenum Press, New York, 1981.
9. Lindgren, B.W. and McElrath, G.W., *Introduction to Probability and Statistics*, The Macmillan Company, New York, 1959.
10. Loe'Ve, M., *Probability Theory and Mathematical Statistics*, John Wiley & Sons, New York, 1963.
11. Mises, R.V., *Probability, Statistics and Truth*, George Allen and Unwin, New York, 1957.
12. Parzen, E., *Modern Probability Theory and its Applications*, John Wiley & Sons, New York, 1960.
13. Rahman, N.A., *Practical Exercises in Probability and Statistics*, Charles Griffin, London, 1972.
14. Rohatgi, V.K., *An Introduction to Probability Theory and Mathematical Statistics*, Wiley Eastern Ltd., Publishers, New Delhi, 1993.
15. Uspensky, J.V., *Introduction to Mathematical Probability*, Tata McGraw Hill, Delhi, 1965.

142. Beta type I and beta type II distributions are same excepting in respect of \_\_\_\_\_.
143. Characteristic function is not useful to work out the moments in case of \_\_\_\_\_ distribution.
144. If a variate  $X \sim \gamma(\alpha, 1)$ , the distribution of  $X$  is same as that of \_\_\_\_\_ distribution.
145. If a variate  $Y \sim b(y; n, p)$ , variate  $Y/n$  is known as \_\_\_\_\_ variate.
146. A variable  $X$  representing the angles and satisfying the condition  $\int_0^\pi f(x) dx = 1$  for

## SECTION-C

## Multiple Choice Questions

Select one correct alternative out of the given ones

- Q. 1 Two random variables  $X$  and  $Y$  are said to be independent if:  
 (a)  $E(XY) = 1$   
 (b)  $E(XY) = 0$   
 (c)  $E(XY) = E(X)E(Y)$   
 (d)  $E(XY) = \text{any constant value}$
- Q. 2 If  $X$  and  $Y$  are two random variables such that their expectations exist and  $P(x \leq y) = 1$ , then  
 (a)  $E(X) \leq E(Y)$   
 (b)  $E(X) \geq E(Y)$   
 (c)  $E(X) = E(Y)$   
 (d) none of the above
- Q. 3 If  $X$  and  $Y$  two random variables and their expected values are  $\bar{X}$  and  $\bar{Y}$  respectively, then  
 (a)  $E\{(X - \bar{X})(Y - \bar{Y})\} = 0$   
 (b)  $E\{(X - \bar{X})(Y - \bar{Y})\} = 1$   
 (c)  $E\{(X - \bar{X})(Y - \bar{Y})\} = C$  (constant)  
 (d) all the above
- Q. 4 If  $X$  is a random variable with its mean  $\bar{X}$ , the expression  $E(X - \bar{X})^2$  represents:  
 (a) the variance of  $X$   
 (b) second central moment  
 (c) both (a) and (b)  
 (d) none of (a) and (b)
- Q. 5 If  $X$  and  $Y$  are two random variables, then  
 (a)  $E\{(XY)^2\} = E(X^2)E(Y^2)$   
 (b)  $E\{(XY)^2\} = E(X^2 Y^2)$   
 (c)  $E\{(XY)^2\} \geq E(X^2)E(Y^2)$   
 (d)  $E\{(XY)^2\} \leq E(X^2)E(Y^2)$
- Q. 6 The outcomes of tossing a coin three times are a variable of the type:  
 (a) continuous random variable  
 (b) discrete random variable  
 (c) neither discrete nor continuous random variable  
 (d) discrete as well as continuous random variable
- Q. 7 The height of persons in a country is a random variable of the type:  
 (a) continuous r.v.  
 (b) discrete r.v.  
 (c) neither discrete nor continuous r.v.  
 (d) continuous as well as discrete r.v.

$f(x) \geq 0$  and  $0 \leq x \leq \pi$  follows \_\_\_\_\_ distribution.

147. The parameters of a multinomial distribution having  $k$  classes and  $n$  observations are \_\_\_\_\_.

148. The variable  $y = -2 \log x$  for  $X \sim U(0, 1)$  is distributed as Chi-square with \_\_\_\_\_ d.f.

149. The variable  $Y = -2 \sum_{i=1}^n \log X_i$ , where  $X_i$  are i.i.d.  $U(0, 1)$ , follows \_\_\_\_\_ distribution.

150. A distribution having always mean and standard deviation equal is \_\_\_\_\_.

Q. 8 If  $X$  and  $Y$  are two random variables with means  $\bar{X}$  and  $\bar{Y}$  respectively, then the expression

$$E\{(X - \bar{X})(Y - \bar{Y})\}$$

is called:

- (a) variance of  $X$   
 (b) variance of  $Y$   
 (c) COV ( $X, Y$ )  
 (d) moments of  $X$  and  $Y$

Q. 9 If  $X$  is a random variable,  $E(e^{itX})$  is known as:

- (a) characteristic function  
 (b) moment generating function  
 (c) probability generating function  
 (d) all the above

Q. 10 If  $X$  is a random variable, the  $E(t^r)$  is known as:

- (a) characteristic function  
 (b) moment generating function  
 (c) probability generating function  
 (d) the  $x^{\text{th}}$  moment

Q. 11 If  $X$  is a random variable with mean  $\mu$ , the  $E(X - \mu)^r$  is called:

- (a) variance  
 (b)  $r^{\text{th}}$  raw moment  
 (c)  $r^{\text{th}}$  central moment  
 (d) none of the above

Q. 12 If  $X$  is a random variable,  $E(e^{itX})$  is known as:

- (a) characteristic function  
 (b) moment generating function  
 (c) probability generating function  
 (d) all the above

Q. 13 If  $X_1, X_2, \dots, X_n$  be a sequence of mutually independent random variables where  $X_i$  can take only positive integral values and

$$S_m = \sum_{i=1}^m X_i \quad (m \leq n), S_n = \sum_{i=1}^n X_i, E(X) = \mu > 0$$

then

(a)  $E\left(\frac{S_m}{S_n}\right) = 1$

(b)  $E\left(\frac{S_m}{S_n}\right) = 0$

(c)  $E\left(\frac{S_m}{S_n}\right) = \frac{m}{n}$

(d)  $E\left(\frac{S_m}{S_n}\right) = \infty$

Q. 14 If  $X$  is a random variable which can take only non-negative values, then

- (a)  $E(X^2) = [E(X)]^2$   
 (b)  $E(X^2) \geq [E(X)]^2$   
 (c)  $E(X^2) \leq [E(X)]^2$   
 (d) none of the above

Q. 15 If  $X$  is a random variable having its p.d.f.  $f(x)$ , the  $E(X)$  is called:

- (a) arithmetic mean  
 (b) geometric mean  
 (c) harmonic mean  
 (d) first quartile

Q. 16 If  $X$  is a random variable and  $f(x)$  is its

p.d.f.,  $E\left(\frac{1}{X}\right)$  is used to find:

- (a) arithmetic mean  
 (b) harmonic mean  
 (c) geometric mean  
 (d) first central moment

Q. 17 If  $X$  is a random variable and its p.d.f. is  $f(x)$ ,  $E(\log x)$  represents:

- (a) arithmetic mean  
 (b) geometric mean  
 (c) harmonic mean  
 (d) logarithmic mean

Q. 18 If  $X$  and  $Y$  are two random variables, the covariance between the variables  $aX + b$  and  $cY + d$  in terms of COV ( $X, Y$ ) is:

- (a) COV ( $aX + b, cY + d$ ) = COV ( $X, Y$ )  
 (b) COV ( $aX + b, cY + d$ ) =  $abcd \times \text{COV} (X, Y)$   
 (c) COV ( $aX + b, cY + d$ ) =  $ac \text{COV} (X, Y) + bd$   
 (d) COV ( $aX + b, cY + d$ ) =  $ac \text{COV} (X, Y)$

Q. 19 If  $X$ ,  $Y$  and  $Z$  are three random variables, then

- (a)  $\text{COV}(X + Y, Z) = \text{COV}(X, Z) + \text{COV}(Y, Z)$
- (b)  $\text{COV}(X + Y, Z) = \text{COV}(X, Z, YZ)$
- (c)  $\text{COV}(X + Y, Z) = \text{COV}(X, Y, Z)$
- (d)  $\text{COV}(X + Y, Z) = 0$

Q. 20 If  $X$  is a random variable and  $r$  is an integer, then  $E(X^r)$  represents:

- (a)  $r^{\text{th}}$  central moment
- (b)  $r^{\text{th}}$  factorial moment
- (c)  $r^{\text{th}}$  raw moment
- (d) none of the above

Q. 21 For Bernoulli distribution with probability  $p$  of a success and  $q$  of a failure, the relation between mean and variance that holds is:

- (a) mean < variance
- (b) mean > variance
- (c) mean = variance
- (d) mean  $\leq$  variance

Q. 22 The outcomes of an experiment classified as success  $A$  or  $\bar{A}$  failure will follow a Bernoulli distribution iff:

- (a)  $P(A) = \frac{1}{2}$
- (b)  $P(A) = 0$
- (c)  $P(A) = 1$
- (d)  $P(A)$  remains constant in all trials

Q. 23 The mean and variance of a binomial distribution are 8 and 4, respectively. Then,  $P(X = 1)$  is equal to:

- (a)  $\frac{1}{2^{12}}$
- (b)  $\frac{1}{2^4}$
- (c)  $\frac{1}{2^6}$
- (d)  $\frac{1}{2^8}$

Q. 24 If for a binomial distribution,  $b(n, p)$ ,  $n = 4$  and also  $P(X = 2) = 3P(X = 3)$ , the value of  $p$  is:

- (a)  $\frac{9}{11}$
- (b) 1
- (c)  $\frac{1}{3}$
- (d) none of the above

Q. 25 If for a binomial distribution  $b(n, p)$ , mean = 4, variance =  $\frac{4}{3}$ , the probability,  $P(X \geq 5)$  is equal to:

- (a)  $\left(\frac{2}{3}\right)^6$
- (b)  $\left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)$
- (c)  $\left(\frac{1}{3}\right)^6$
- (d)  $4\left(\frac{2}{3}\right)^6$

Q. 26 An experiment succeeds twice as often as it fails. The chance that in the next six trials, there shall be at least four successes is:

- (a)  $\frac{240}{729}$
- (b)  $\frac{489}{729}$
- (c)  $\frac{496}{729}$
- (d) none of the above

Q. 27 A manufacturer produces switches and experiences that 2 per cent switches are defective. The probability that in a box of 50 switches, there are at most two defective is:

- (a)  $2.5 e^{-1}$
- (b)  $e^{-1}$
- (c)  $2 e^{-1}$
- (d) none of the above

Q. 28 If  $X \sim b\left(3, \frac{1}{2}\right)$  and  $Y \sim b\left(5, \frac{1}{2}\right)$ , the probability of  $P(X + Y = 3)$  is:

- (a)  $7/16$
- (b)  $7/32$
- (c)  $11/16$
- (d) none of the above

Q. 29 If  $X$  and  $Y$  are two Poisson variates such  $X \sim P(1)$  and  $Y \sim P(2)$ , the probability,  $P(X + Y < 3)$  is:

- (a)  $e^{-3}$
- (b)  $3e^{-3}$
- (c)  $4e^{-3}$
- (d)  $8.5e^{-3}$

Q. 30 If  $X$  is a binomial variate with its mean  $\mu = 4$  and third moment  $\mu_3 = 4.8$ , the value of Pearson's constant  $\beta_1$  is:

- (a)  $2/3$
- (b)  $5/6$
- (c) 0
- (d) none of the above

Q. 31 If  $X \sim b(n, p)$ , the distribution of  $Y = (n - X)$  is:

- (a)  $b(n, 1)$
- (b)  $b(n, x)$
- (c)  $b(n, p)$
- (d)  $b(n, q)$

Q. 32 A family of parametric distribution in which mean is equal to variance is:

- (a) binomial distribution
- (b) gamma distribution
- (c) normal distribution
- (d) Poisson distribution

Q. 33 A family of parametric distributions in which mean is always greater than its variance is:

- (a) binomial distribution
- (b) geometric distribution
- (c) both (a) and (b)
- (d) neither (a) nor (b)

Q. 34 A family of parametric distributions having mean  $<, =, >$  variance is:

- (a) gamma distribution
- (b) exponential distribution
- (c) logistic distribution

(d) all the above

Q. 35 The family of parametric distributions which has mean always less than variance is:

- (a) beta distribution
- (b) lognormal distribution
- (c) Weibull distribution
- (d) negative binomial distribution

Q. 36 The family of parametric distributions, for which the mean and variance does not exist, is:

- (a) Polya's distribution
- (b) Cauchy distribution
- (c) negative binomial distribution
- (d) normal distribution

Q. 37 A family of parametric distributions which has mean always less than its variance is:

- (a) negative binomial distribution
- (b) Polya's distribution
- (c) beta distribution of second kind
- (d) hypergeometric distribution

Q. 38 The family of parametric distributions for which moment generating function does not exist is:

- (a) student  $t$ -distribution
- (b) Pareto distribution
- (c)  $F$ -distribution
- (d) all the above

Q. 39 The distribution possessing the memoryless property is:

- (a) gamma distribution
- (b) geometric distribution
- (c) hypergeometric distribution
- (d) all the above

Q. 40 The distribution in which the probability at each successive draw varies is:

- (a) hypergeometric distribution
- (b) geometric distribution
- (c) binomial distribution
- (d) discrete uniform distribution

Q. 41 In hypergeometric distribution, H.G. ( $N$ ,

$k, n$ ), if  $N \rightarrow \infty, \frac{k}{N} \rightarrow p$ , the hypergeo-

metric distribution reduces to:  
(a) binomial distribution

- (a) Bernoulli variate  
(b) Poisson variate  
(c) binomial variate  
(d) negative binomial variate

Q. 209 If a variable  $X$  has the p.d.f.,  $f(x) = \frac{1}{4} \cdot x e^{-x/2}$  for  $x > 0$ , then the variable  $X$  is distributed as:

- (a) gamma variate  
(b) chi-square variate  
(c) both (a) and (b)  
(d) neither (a) nor (b)

Q. 210 If a variable  $X$  has the p.d.f.,  $f(x) = \frac{1}{4} \cdot x e^{-x/2}$  for  $0 \leq x < \infty$ , then the distribution has mean and variance as:

- (a) mean = 2, variance = 4  
(b) mean = 1/2, variance = 1/4  
(c) mean = 4, variance = 2  
(d) mean = 4, variance = 8

## ANSWERS

### SECTION-B

(1) finite (2) domain (3)  $x : 0 \quad 1 \quad 2 \quad 3$   
 $p(x) : \frac{1}{8} \quad \frac{3}{8} \quad \frac{3}{8} \quad \frac{1}{8}$

(4)  $z : 0 \quad 1 \quad 2 \quad 3 \quad 4$   
 $p(z) : \frac{1}{8} \quad \frac{3}{8} \quad \frac{1}{8} \quad \frac{2}{8} \quad \frac{1}{8}$  (5)  $\frac{1}{26}$  and  $E(x) = 3.96$

(6)  $\frac{3}{4}$  (7) 1 (8) unity (9) (a) 0.1 (b) 0.5 (c) 2.85

(10) (a)  $\frac{5}{12}$  (b)  $\frac{2}{11}$  (11) p.d.f (12) (a)  $E(X) =$

$\int_a^b x f(x) dx$  (b)  $\log G = \int_a^b \log x f(x) dx$  (c)  $\frac{1}{H} =$

$\int_a^b \frac{1}{x} f(x) dx$  (d)  $E(X^2) = \int_a^b x^2 f(x) dx$  (e)  $\mu_2 =$

$\int_a^b x^2 f(x) dx - \left\{ \int_a^b x f(x) dx \right\}^2$  (f) M.D.

$\int_a^b |x - \mu| f(x) dx$  (g)  $\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$

(h)  $\int_a^b f(x) dx = \frac{i}{4}$  (i)  $\int_a^b f(x) dx = \frac{i}{10}$  (j)

$\int_a^b f(x) dx = \frac{i}{100}$  (13) 0.4 (14) (a) 1 (b) 1 (c) 2 (d)

(15) 1 (16)  $\frac{3}{16}(y-3)$  (17)  $\frac{1}{4\sqrt{1-y^2}}$  (18)  $\frac{1}{2}$  (19)

(20)  $np; npq$  (21) greater than (22)  $\frac{5}{16}$  (23)  $\frac{22}{72}$

(24)  $\frac{27}{8}e^{-3}$  (25)  $\frac{63}{512}$  (26) geometric (27) relate

(28) lack of memory (29) 1 (30) greater than (31)  $\lambda$

= 1 (32)  $\beta = 1$  (33) dependent (34) hypergeometric

distribution (35) fixed (36)  $N \rightarrow \infty$  and  $\frac{k}{N} \rightarrow p$

(37) equal (38) symmetrical (39) does not exist (40)

$\frac{b-a}{4}$  (41)  $\frac{(b-a)^2}{12}$  (42) exist (43) (a)  $\frac{\theta}{2}$  (b)  $\frac{\theta^2}{12}$

(c) 0 (d)  $\frac{\theta^4}{80}$  (e) 0 (f)  $\frac{9}{5}$  (44) one (45) symmetrical

(46) more (47)  $U(0, 1)$  (48) twice (49) leptokurtic

(50)  $\frac{1}{\lambda} \log 2$  (51)  $N(0, 1)$  (52)  $X =$

$\frac{1}{\sigma\sqrt{2\pi}} e^{-1/2}$  (54)  $N(n, \mu, n\sigma^2)$  (55)  $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

(56) (a) unimodal (b) not skewed (c) Mesokurtic

(57) 10 : 12 : 15 (58)  $\sigma\sqrt{\frac{2}{\pi}}$  or  $\frac{4}{5}\sigma$  (59) zero (60)

$e^{\mu + \frac{1}{2}\sigma^2 t^2}$  (61)  $e^{i\mu t - \frac{1}{2}\sigma^2 t^2}$  (62)  $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

(63)  $N(21.0, 18.5)$  (64) (a) 0.62% (b) 10.56% (c) 38.29% (d) 24.17% (e) 29.63% (65) (a) 3 (b) 53

(c) 191 (d) 121 (e) 148 (66)  $e^{\mu + \frac{1}{2}\sigma^2}$  (67)  $e^{2\mu + \sigma^2}$  (68) elongated (69) more (70) lognormal

(71)  $\log - N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$

(72)  $\log - N\left(\mu, \frac{\sigma^2}{n}\right)$

(73)  $\frac{1}{\sqrt{5\pi}} \left\{ 1 + \left( \frac{X-2}{5} \right)^2 \right\}$

(74) not existing (75) not existing (76) non-existing

(77)  $X = 5$  (78) No (79)  $X = 2$  (80)  $C(\alpha, \beta)$

(81)  $C(0, 1)$  (82) are not (83)  $\alpha = 1, \beta = 1$  (84)

$\frac{\alpha}{\alpha + \beta}$  (85)  $\frac{\alpha}{\beta - 1}$  (86)  $\alpha$  and  $\beta$  (87) mean =  $\alpha \times$

variance (88) positive skewness (89) leptokurtic (90)

$Y(n_1 + n_2)$  (91)  $\beta_1(n_1, n_2)$  (92) incomplete gamma function

(93)  $\alpha$  (94) symmetrical (95) modelling problem

(96) reliability theory (97) corrosion (98)  $(0.8)^{25}$

(99) binomial;  $n$  and  $q$  (100)  $4e^{-2}$  (101) 2 (102)  $\sigma =$

12.2 (103) 4 (104) hyperbolic (105)  $\mu_3^2 = 14\mu_2^2$  (106)

$\mu_3 = 4(\mu_2 + 8\mu_1)$  (107) gamma distribution (108)

(a)  $\beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$  (b)  $F$ -distribution with d.f.  $(n_1, n_2)$

(c)  $\chi^2$  will d.f.  $(n_1 + n_2)$  (d)  $\beta_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$  (e)  $\chi^2$  with

$(n_1 - n_2)$  d.f. (109) half (110) four times (111)

$\left(\frac{2}{n}\right)^{1/2}$  (112) leptokurtic (113) normal distribution

(114) Chi (115)  $n$  (116) non-central chi-square (117) Chi-square with  $(n-1)$  d.f. (118) Chi-square with

(126) least (127)  $t$ -test (128)  $\chi^2$  test (129)  $F$ -test

(130)  $\chi^2$ -test (131) Fisher's  $t$  (132) Student's  $t$

(133)  $F = t^2$  (134)  $\frac{1}{2} \left( \frac{1}{v_2} - \frac{1}{v_1} \right)$  (135) less than

(136) does not exist (137)  $F_{\alpha; v_1, v_2} = \frac{1}{F_{1-\alpha; v_2, v_1}}$

(138)  $v_1 F = \chi^2$  (139) normal (140)  $\chi^2$  with  $(n-1)$

d.f. (141)  $N(0, 1)$  (142) range (143) hypergeometric

(144) exponential (145) Pseudo binomial (146) circular distribution (147)  $n, p_1, p_2, \dots, p_k$  (148)  $n$

(149) chi-square (150) exponential distribution

SECTION-C

(1) c (2) a (3) a (4) c (5) d (6) b

(7) a (8) c (9) b (10) c (11) c (12) a

(13) c (14) c (15) a (16) b (17) b (18) d

(19) a (20) c (21) b (22) d (23) a (24) c

(25) d (26) c (27) a (28) b (29) d (30) b

(31) d (32) d (33) c (34) d (35) d (36) b

(37) a (38) d (39) b (40) a (41) a (42) d

(43) c (44) b (45) a (46) c (47) b (48) c

(49) d (50) a (51) b (52) c (53) c (54) b

(55) a (56) c (57) a (58) c (59) b (60) c

(61) d (62) d (63) c (64) d (65) a (66) b

(67) d (68) c (69) d (70) c (71) d (72) b